Elec Eng Complex Calculation R47 & C47 Appnote 001a 2026-01-09

Appnote 001a updates appnote 001 with a new R47 section, presenting a worked electrical design example demonstrating the use of aspects of complex-number and engineering functions on the R47 and C47 calculators.

The tutorial example, is to size an XLPE aluminium cable connecting a neutral earthing transformer to the 33 kV terminals of a power transformer. As the neutral transformer carries no normal load current, the cable must be designed to withstand the thermal effects of a three-phase short circuit at the cable box within the specified fault clearance time.

The calculation sequence determines the system source impedance, refers it to the transformer secondary side, and combines it with the transformer impedance to establish the maximum prospective three-phase fault current.

The example highlights typical calculator engineering settings, demonstrates practical use of complex arithmetic in both polar and rectangular form, and applies the standard adiabatic short-circuit thermal equation to estimate the minimum conductor cross-sectional area, which is then checked against standard cable sizes to complete the design.

This appnote refers to the original problem on p3 of my original 2019 WP34C appnote https://cocoon-creations.com/download/WP34C_Complex_Description_Jaco.pdf.

The problem:

Calculate the XLPE cable size required to connect a neutral transformer (NEC) close to the 33 kV terminals of a 80 MVA 132/33 kV transformer. There is no normal load current in the neutral transformer, but the cable must withstand a large 3 phase short circuit in the cable box. The main power transformer is connected to a power system with a 132 kV fault level of 16.5 kA @ X/R = 10. The backup transformer protection is set to guarantee fault clearance in 0.5 s:

The solutions illustrating various learning outcomes for C47 and R47 below:

R47, typical engineering settings:

Set display format to SIG 3 to maintain 3 digits after the radix, eg. 1.234, a total of 4 digits	DISP F5 (SIG) 3
Set complex format to the form 1.234+*j*·4.567	↑ F2 (CPX *j*) F4 (CPXmul) F6 (MULT·)
Set angle display mode ADM to DEG	TRG F1 (DEG)
Set complex output display format to rectangular	CPX F5 (RECT)

Calculate the 132 kV system source impedance from the specified fault level (16.5 kA @ 132 kV with X/R=10).	$Z_{3ph} = (V / \sqrt{3}) / I_{fault}$	1 3 2 EEX 3 ENTER 3 √x ÷ 1 6 . 5 EEX 3 ÷ 1 0 ATAN COMPLEX 4.619∡84.29° Ω Shown in polar notation, in DEG, because the input to COMPLEX, was an angle in DEG.
Refer the source impedance to the 33 kV side to later calculate the 33 kV maximum current.	$Z_{33kV} = Z_{132kV} / n^2$ $n = V_{primary} / V_{secondary}$	1 3 2 ENTER 3 3 ÷ x² ÷ STO α ZS 3 3 ENTER 'zs33': 0.02872+j·0.2872Ω Calculation results in rectangular notation because of RECT setting.
Calculate the impedance of the 132/33 kV transformer, 80 MVA, 11%, X/R=28.67, referred to 33 kV side.	$Z_{t33} = (Z\% / 100) \times (V^2 / S)$	3 3 EEX 3 x² 8 0 EEX 6 ÷ 1 1 % 2 8 . 6 7 ATAN COMPLEX STO α zt 3 3 ENTER 'zt33': 1.497∡88.00° Ω Shown in polar notation, in DEG, because input to COMPLEX, was angle in DEG.
Calculate the 3ph fault current at the 33 kV side of the transformer	$I = (V / \sqrt{3}) / (Z_s + Z_t)$	RCL α ZS 3 3 ENTER + 3 3 EEX 3 x⇄y ÷ 3 √x ÷ |x| 10 670 A
Estimate the minimum surface area for Alu cable for 0.5 second fault clearance time.	$A = (I^2 t)^{0.5} / 94$ (source)	x² . 5 × √x 9 4 ÷ 80.27: Cross section area > 80.27 mm². The standard cable size is 95 mm²

Notes:

1. Once a number is displayed in complex notation, >P and >R commands are for display purposes only.
2. Calculations will be correct whether in polar or rectangular form.
3. Polar and rectangular formats can be mixed at will. Internal storage is always in rectangular.
4. A tagged angle as an input to COMPLEX will be transferred to the complex angle and will be displayed in polar mode in the same tagged ADM.
5. The EE example was by choice not done in the normal per unit (p.u.) system to be able to illustrate the SIG 3 functionality, which in the percentage system or p.u. system can trivially be done in FIX 3.

The problem redone on C47, using the PFX (Prefix) menu with multipliers .k, .M, etc., also illustrating the >R >P sequence of the HP.RP[] setting.

C47 using COMPLEX numbers Using Prefix, →R, →P			**C47 using REAL numbers Using Prefix, →R, →P**		
	Using COMPLEX with →R →P with HP.RP (clear)	**Stack**	**Using →R →P with HP.RP (clear)**	**Stack**	**Comments**
Defaults assumed, i.e. eRPN, DEG, RECT	f[MODE] HP.RP[] (clear) f[DISP] [SIG] 03	0.000 0.000 0.000 0.000	f[MODE] HP.RP[] (clear) f[MODE] SIG 03	0.000 0.000 0.000 0.000	Clear HP.RP to use natural sequence for COMPLEX, →R, →P
1. Calculate the 132 kV source impedance, given the specified fault level is 16.5 kA with X/R=10. Z = V / I	132000 ENTER 16500 ÷ 3 √x ÷ 10 f[ATAN]	0.000 0.000 4.619 84.29°	10 f[ATAN] 132000 ENTER 16500 ÷ 3 √x ÷	0.000 0.000 84.29° 4.619	Source impedance = 4.619 Ω ∠ 84.29°
→R →P COMPLEX	g[→R] Change 2xReals to Rect ! f[COMPLEX] g[→P] Change X to Polar !	0.000 0.000 0.000 4.619∡84.29°			
2. Refer the source impedance to the 33 kV side in order to later calculate the 33 kV maximum current.	33 ENTER 132 ÷ x^2 ×	0.000 0.000 0.000 0.02872+0.2872*i*	33 ENTER 132 ÷ x^2 × x⇄y g[→R] Change 2xReals to Rect !	0.000 0.000 Re = 0.02872 Im = 0.2872	Natural entry sequence: r, θ Source impedance = 0.02872 + i.0.2872 Ω

3. Calculate the impedance of the 132/33 kV transformer, 80 MVA, 11%, X/R=28.67, referred to 33 kV side. S = VI Z = V / I	g[HOME] g[Prefix] 80[•M] 33[•k] ÷ 3√x ÷ 0.11 ÷ 33[•k] 3√x ÷ x⇄y ÷ 28.67 f[ATAN] g[→R] Change 2xReals to Rect ! f[COMPLEX]	0.000 0.000 0.02872+0.2872*i* 0.05220+1.496*i*	28.67 f[ATAN] fff g[Prefix] 80[•M] 33[•k] ÷ 3√x ÷ 0.11 ÷ 33[•k] 3√x ÷ x⇄y ÷	0.02872 0.2872 88.00° 1.497	Reach **HOME** by fff or g[HOME] Calc Z angle Calc Z Tx, Tx Impedance = 1.497 ∠ 88.00° Ω
4. Calculate the current at the 33 kV side of the transformer I = V / ΣZ	+ 33 [•k] 3 √x ÷ x⇄y ÷	0.000 0.000 0.000 483.6+10660.*i*	x⇄y g[→R] Change 2xReals to Rect ! RCL+ Z x⇄y RCL+ T x⇄y g[→P] Change 2xReals to Polar ! x⇄y 33 [•k] 3 √x ÷ x⇄y ÷	0.02872 0.2872 87.40° 10670 Current is 10670 ∠ -87.40° A	Add impedances
5. Estimate the cross-section area for Aluminium cable, assuming 0.5-second fault clearance time. $t = k^2 A^2 / I^2$	f [|x|] f x^2 0.5 × √x 94 ÷	0.000 0.000 0.000 80.27 Minimum size is 80.27 mm^2, and the cable must be the next size up, i.e. 95 mm^2 cross sectional area.	f x^2 0.5 × √x 94 ÷	0.02872 0.2872 87.40° 80.27 Minimum size is 80.27 mm^2, and the cable must be the next size up, i.e. 95 mm^2 cross sectional area.	Ignore angle, use scalar current Calculate minimum area

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